Total No. of Printed Pages : 6]

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015267 & 015268

B. Sc. (Hons) Semester IV Open Book Examination 2021 – 22

CHEMISTRY

Paper No. CHB-401 : Inorganic Chemistry–II and Organic Chemistry–III

Time : 4 hours 30 minutes **Full Marks : 70**

Instructions :

(i) The Question Paper contains 08 questions out of which you are required to answer any 2 questions each from Sections A and B. The question paper is of 70 Marks with each question carrying $17 \cdot 5$ Marks.

(ii) The total duration of the examination will be 4:30 Hours (Four Hours and Thirty Minutes), which includes the time for downloading the question paper from the portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.

(iii) For the students with benchmark disability as per Persons with Disability Act, the total duration of examination shall be 6 Hours (Six Hours) to complete the examination process, which includes the time for downloading the question paper from the portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.

(iv) Answers should be hand-written on plain white A4 size paper using black or blue pen. Each question can be answered in up to 350 words on 3 (Three) plain A4 size paper (only one side to be used).

(v) Answers to each question should start from a fresh page. All pages are required to be numbered. You should write your Course Name, Semester, Examination Roll Number, Paper Code, Paper Title, Date and Time of Examination on the first sheet used for answers.

(vi) Use separate Answer-Sheets for each Section.

Section—A
(Inorganic Chemistry–II)

Q.1. 1. Giving reasons arrange the following in order of their increasing acid strength :

$$H_3MnO_4, HMnO_4, H_2MnO_4$$

3

2. Write conjugate acids and bases of H_2O, HCO_3^- and HSO_4^-. **3**

3. Describe Lux-Flood concept of acids and bases. Complete the reactions and identify the Lux-Flood acids and bases in the following : **5**

(i) $Na_2O + ZnO \longrightarrow$

(ii) $FeO + SiO_2 \longrightarrow$

(iii) $CaO + CO_2 \longrightarrow$

4. State the properties that determine the utility of a solvent. What happens when ammonolysis of $SnCl_4$ takes place? $4\frac{1}{2}$

5. Write autoionization reactions of liquid SO_2 and liquid N_2O_4. **2**

Q.2. 1. Why do the solutions of alkali metals in liquid ammonia behave as very good reducing agents? **2**

2. What is HSAB principle? Predict the preferred directions of the following reactions. Give reasons also : **4**

   (i) $HgF_2 + BeI_2 \rightleftharpoons HgI_2 + BeF_2$

   (ii) $TiI_4 + 2TiF_2 \rightleftharpoons TiF_4 + 2TiI_2$

3. Write protolysis reactions of urea, acetamide and sulphamic acid in liquid ammonia. $4\frac{1}{2}$

4. In each of the following pairs which would you expect to be a better Lewis acid? Explain : **3**

   (i) BH_3 or $B(CH_3)_3$

   (ii) BF_3 or BBr_3

5. What do you understand by differentiating solvent? What happens when HNO_3 is dissolved in $HClO_4$ and H_2SO_4? **4**

Q.3. 1. "Transition metal elements have variable oxidation states." Justify the statement with suitable examples. $2\frac{1}{2}$

2. What are the primary and secondary valencies according to Werner's theory? Predict the primary and secondary valency in $[Co(NH_3)_5Cl]Cl_2$ complex. **3**

3. Calculate the effective atomic number of the metal in the following : **3**

   (i) $[Fe(CO)_5]$

   (ii) $[Co(NH_3)_6]Cl_3$

   (iii) $K_4[Fe(CN)_6]$

4. Predict the Geometry, shape and hybridization in the following coordination complexes : **3×3**

   (i) $[Co(NH_3)_6]^{3+}$

   (ii) $[Cu(NH_3)_4]SO_4 \cdot H_2O$

   (iii) $[NiCl_4]^{2-}$

Q.4. 1. Write the IUPAC name of the following coordination compounds : **3×1**

   (i) $[Co(SO_4)(NH_3)_5]NO_3$

   (ii) $[Cr(en)_2Cl_2]$

   (iii) $[(NH_3)_5Co\text{-}NH_2\text{-}Co(NH_3)_5](NO_3)_5$

2. Write the difference between coordination and ionization isomers using suitable

examples. Which type of isomerism are possible in $[Co(en)_2NO_2 \cdot Cl]SCN$ and $[Co(en)_2NO_2 \cdot SCN]Cl$ compounds? **3**

3. The size from Ce to Lu decreases. Justify. **2**

4. What are the hybridization and shape of $[ZnCl_4]^{2-}$? Why does it dissociate into $[ZnCl_3]^-$ and Cl^- ions? $2\frac{1}{2}$

5. Write the outer electronic configuration of Nd and Dy in ground state. Also predict the number of unpaired electrons in Nd^{3+} and Dy^{3+}. **3**

6. What is the most common oxidation state for the lanthanides? Discuss the variable oxidation states in the lanthanides with suitable examples. **4**

Section—B
(Organic Chemistry–III)

Q.5. 1. The following statement is True or False? Justify your answer :

"Fructose is having eight D-stereoisomers."

2

2. How do you prepare following compound from ethyl acetoacetate? Write down the stepwise mechanism :

(Structural framework of 2,4-dimethyl-3-acetylpyrrole execution matrix)

4

3. Explain why α-position of naphthalene is more reactive towards electrophile but Friedel-Crafts alkylation (with CH_3-CH_2-CH_2-Br/$AlCl_3$) happens at β-position? **4**

4. Predict the structure of the products (**A** to **E**) from the following reactions : $7\frac{1}{2}$

(i) Naphthalene $\xrightarrow{Na_2Cr_2O_7/H_2SO_4}$ **A** $\xrightarrow{\text{Butadiene}}$ **B** $\xrightarrow[\text{2. Zn dust}]{\text{1. } CrO_3}$ **C**

(ii) *(Indigotin derivative structural core)* $\xrightarrow{\text{aq. NaOH}}$ **D** + **E**

Q.6. 1. The following statement is True or False? Justify your answer.

"Pyridine readily undergoes nitration with conc. H_2SO_4 and fuming HNO_3."

2

2. How do you prepare following compound from toluene? Write down the stepwise mechanism :

(Structural framework layout of 1-methyl-4-ethylnaphthalene platform)

4

3. Explain why 3-hydroxypyridine shows phenol character but 2-hydroxypyridine does not. **4**

4. Predict the structure of the products (**A** to **E**) from the following reactions : $7\frac{1}{2}$

(i) Succinaldehyde $\xrightarrow{NH_3, \Delta}$ **A** $\xrightarrow{\text{Solid KOH}}$ **B** $\xrightarrow{CHCl_3, EtONa}$ **C**

$$\begin{array}{l} \quad CHO \\ \quad\; | \\ (CHOH)_3 \xrightarrow{4HIO_4} \mathbf{D} + \mathbf{E} \\ \quad\; | \\ \; CH_2OH \end{array}$$

(ii)

Q.7. 1. The following statement is True or False? Justify your answer.

"Reduced form of indigotin is blue colour."

2

2. How do you prepare following compounds from erythrose? Write down the stepwise mechanism : **4**

   (i) D-arabinose

   (ii) D-fructose

3. Explain why thiophene undergoes nitration with fuming nitric acid but furan does not. **4**

4. Predict the structure of the products (**A** to **E**) from the following reactions : $7\frac{1}{2}$

   (i) 2Benzene $\xrightarrow{AlCl_3}$ **A** $\xrightarrow{H_2CrO_4}$ **B**

   (ii) Ph-CH=CH-COOH $\xrightarrow[\text{2. Zn}]{\text{1. } H_2SO_4, \Delta}$ **C** $\xrightarrow{O_2}$ **D** $\xrightarrow[ZnCl_2, \Delta]{\text{2 resorcinol}}$ **E**

Q.8. 1. The following statement is True or False? Justify your answer.

"Benzene is more aromatic than Naphthalene."

2

2. How do you prepare following compounds from furan? Write down the stepwise mechanism :

   (Conversion cascade: Furan-2-carboxylic acid to Furfuryl alcohol constructs)

   4

3. How do you confirm that glucose exists in cyclic form by mutarotation? **4**

4. Predict the structure of the products (**A** to **E**) from the following reactions : $7\frac{1}{2}$

   (i) $$\begin{array}{l} CH_2OH \\ \quad | \\ \; C{=}O \\ \quad | \\ (CHOH)_2 \xrightarrow[\Delta]{\text{conc. } HNO_3} \mathbf{A} + \mathbf{B} + \mathbf{C} \\ \quad | \\ CH_2OH \end{array}$$

   (ii) Pyridine $\xrightarrow[300\ ^\circ C]{CH_3Br}$ **D** $\xrightarrow{\text{conc. } HNO_3/\text{conc. } H_2SO_4}$ **E**

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ExternalPortals&AcademicResources:

- **ExaminationPortal:** `https://bhu-exam.vercel.app/`
- **StudentResourceCenter:** `https://quashan-me.vercel.app/`
- **AcademicPortfolio:** `https://me-aryan.vercel.app`